

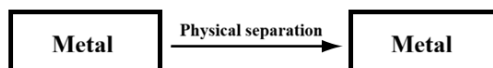
Intensive notes (Topic 3: Metals)

Uses of metals

Metal	Common ore	Uses	Reason
Iron	Haematite (Fe_2O_3)	Construction, car / ship bodies	Hard, strong, malleable, cheap
Copper	Copper pyrite (CuFeS_2)	Electric wire	Good electrical conductor, ductile
		Water pipe	Corrosion resistant, non-toxic
		Cooking utensil	Good heat conductor, non-toxic, corrosion resistant
Aluminium	Bauxite (Al_2O_3)	Overhead power cable	Good electrical conductor, ductile
		Aircraft bodies	Strong, corrosion resistant, low density
		Kitchen foil	Non-toxic, corrosion resistant, malleable
		Soft drink can	Non-toxic, corrosion resistant
		Window frame	Strong, corrosion resistant
Gold	Gold ore (Au)	Jewellery	Corrosion resistant, attractive appearance
Mercury	Cinnabar (HgS)	Thermometer	Expand on heating, liquid at room conditions
Silver	Argentite (Ag_2S)	Jewellery	Corrosion resistant, attractive appearance
		Electronic component	Best electrical conductor, ductile, corrosion resistant
Lead	Galena (PbS)	Make fishing sinkers	High density
Zinc	Zinc sulphide (ZnS)	Galvanizing iron	Forms a protective layer that prevents corrosion
Titanium	---	Supersonic aircraft bodies	Low density, strong, high melting point
		Metal implants in human body	Low density, corrosion resistant, biocompatible

Extraction of metals

1. Extraction of gold, platinum (very unreactive metals)

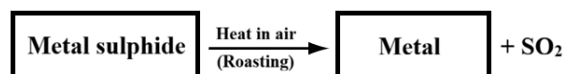


2. Extraction of silver, mercury (unreactive metals)

From metal oxide:

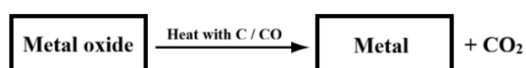


From metal sulphide:

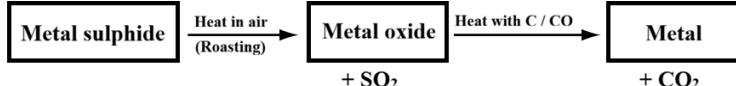


3. Extraction of copper, lead, iron, zinc (more reactive metals)

From metal oxide:

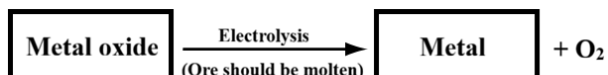


From metal sulphide:

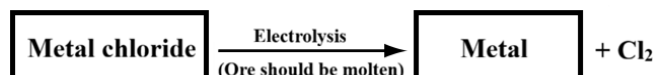


4. Extraction of aluminium, magnesium, calcium, sodium, potassium (very reactive metals)

From metal oxide:

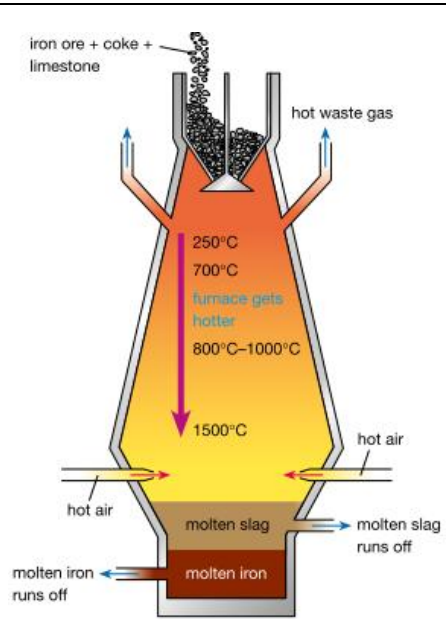


From metal chloride:



Intensive notes (Topic 3: Metals)

Extraction of iron

Step 1 Word equation: Chemical equation:	Hot air is blasted near the bottom of the furnace. Coke (carbon) will react with oxygen in air $\text{Carbon} + \text{Oxygen} \rightarrow \text{Carbon dioxide}$ $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$	
Step 2 Word equation: Chemical equation:	Coke will further react with the carbon dioxide produced in step 1 to form carbon monoxide $\text{Carbon} + \text{Carbon dioxide} \rightarrow \text{Carbon monoxide}$ $\text{C} + \text{CO}_2 \rightarrow 2\text{CO}$	
Step 3 Word equation: Chemical equation:	Carbon monoxide will react with haematite (Fe_2O_3). Iron metal can be obtained in this step. Fe_2O_3 is reduced to Fe in this step. The reducing agent used is carbon monoxide. $\text{Carbon monoxide} + \text{Iron(III) oxide} \rightarrow \text{Carbon dioxide} + \text{Iron}$ $3\text{CO} + \text{Fe}_2\text{O}_3 \rightarrow 3\text{CO}_2 + 2\text{Fe}$	

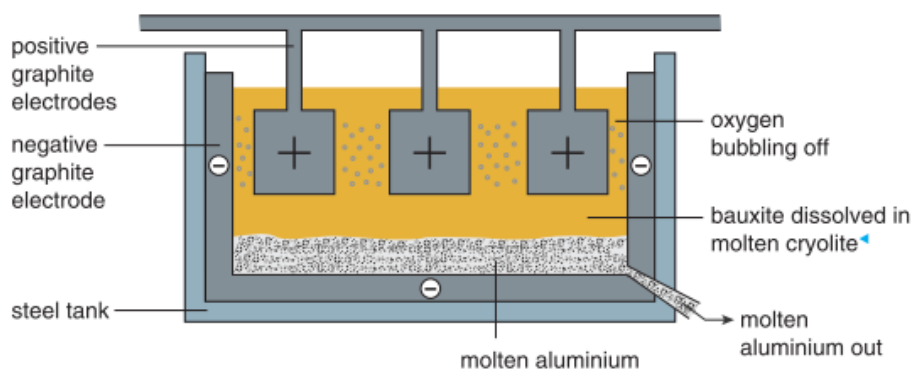
- Limestone is also added to the blast furnace. It is decomposed upon heating to form calcium oxide and carbon dioxide. Calcium oxide can react with the impurities such as silicon dioxide to form slag and it is further removed.

Word equation	Chemical equation
Calcium carbonate $\rightarrow$ Calcium oxide + Carbon dioxide	$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
Calcium oxide + Silicon dioxide $\rightarrow$ Calcium silicate	$\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$

Extraction of aluminium

Raw materials required:

1. Bauxite (Source of aluminium ore containing mainly Al_2O_3)
2. Cryolite (To reduce the melting point of aluminium oxide)



Half equation for the change at anode (+ve electrode)	Half equation for the change at cathode (-ve electrode)
$2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-$	$\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$

- Aluminium oxide must be in molten form. The melting point of bauxite is very high (2000 °C). Therefore the bauxite is dissolved in molten cryolite (a less common aluminium ore) so that the melting point of the mixture of aluminium ore will be reduced to below 2000 °C.

Appearance of metals

Copper	Gold	Mercury	Others you have learnt in this chapter
Reddish brown solid	Golden yellow solid	Silvery grey liquid	Silvery grey solid

Relation between metal reactivity series and the extraction method

- The order of discovery of metals is related to the ease of the extraction of metals.
- Reactive metals (element) form stable ore (compound).
 → The more reactive a metal is, the extraction method of the metal is more difficult.

Q: Explain why iron was discovered and used much earlier than aluminium

A: Iron is less reactive than aluminium. Ores of iron are less stable. Therefore iron can be extracted more easily.

Q: Explain whether carbon can be used to extract magnesium from magnesium oxide

A: No. MgO is very stable / Mg is too reactive. It can only be extracted by electrolysis of molten MgO.

Factors affecting availability of metals

1. Abundance of the metal in Earth's crust
2. Ease and cost of mining the ore
3. Ease and cost of extracting the metal from its ore

In general, the more easily a metal can be extracted, the earlier it was discovered in history.

Ways of conserving metals

Conserving metals is important as metals are non-renewable resources (limited in amount).

1. Reusing metal articles

2. Reducing the use of metals

3. Recycling used metals

Benefits: Conserves metal ore resources, saves energy and other resources, reduces metal wastes and land used for waste disposal, reduces pollution.

Problems: Difficult and time-consuming to sort out metal wastes, cost of collecting, sorting and transporting metal wastes is high, cost of melting and purifying metals is high.

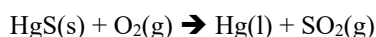
Extractions of metals

1. Heating metal ore alone

A. Metal oxide → Metal + Oxygen Ag, Hg



B. Metal sulphide + Oxygen → Metal + Sulphur dioxide Ag, Hg, (Cu)

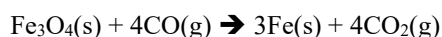


2. Heating metal oxide with reducing agent (e.g. Carbon / carbon monoxide / hydrogen)

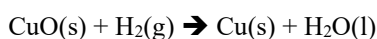
A. Metal oxide + Carbon → Metal + Carbon dioxide Cu, Pb, Fe, Zn



B. Metal oxide + Carbon monoxide → Metal + Carbon dioxide Cu, Pb, Fe, Zn

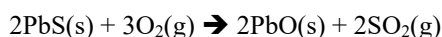


C. Metal oxide + Hydrogen → Metal + Water Cu, Pb, Fe, Zn



*The ores of some metals are in sulphide form. These metal sulphides can be converted to oxide by heating in air

Metal sulphide + Oxygen → Metal oxide + Sulphur dioxide



3. Electrolysis of molten ore

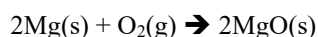
$2\text{Al}_2\text{O}_3\text{(l)} \rightarrow 4\text{Al(l)} + 3\text{O}_2\text{(g)}$ Al, Mg, Ca, Na, K

*Due to high temperature, aluminium metal will be produced in liquid state

Reactions of metals

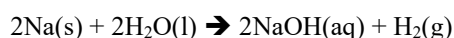
1. Reaction of metal with oxygen

Metal + Oxygen → Metal oxide K, Na, Ca, Mg, Al, Zn, Fe, Pb, Cu, Hg

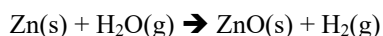


2. Reaction of metal with water / steam

Metal + Water → Metal hydroxide + Hydrogen K, Na, Ca



Metal + Steam → Metal oxide + Hydrogen K, Na, Ca, Mg, Al, Zn, Fe

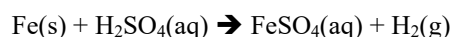


3. Reaction of metal with acid

Metal + Dilute hydrochloric acid → Metal chloride + Hydrogen K, Na, Ca, Mg, Al, Zn, Fe, Pb

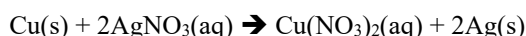


Metal + Dilute sulphuric acid → Metal sulphate + Hydrogen K, Na, Ca, Mg, Al, Zn, Fe, Pb,



4. Displacement reaction

Metal A + Metal B ion → Metal B + Metal A ion A should be more reactive than B.



Intensive notes (Topic 3: Metals)

Metal reactivity series

Most reactive metal

Least reactive metal

(Easy to react to form compounds)

(Difficult to react to form compounds)

K	Na	Ca	Mg	Al	Zn	Fe	Pb	Cu	Hg	Ag	Pt	Au
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Forming most stable ore

Forming least stable ore

(Difficult to extract)

(Easy to extract)

*K is said to be more reactive than Na as K loses electrons more readily than Na

Reaction of metals with oxygen in air

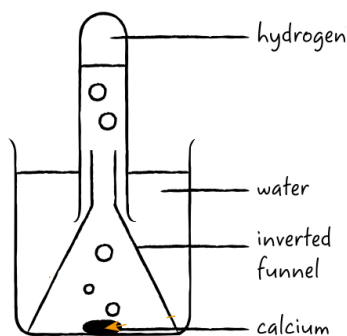
Metal	Observation	Word equation / chemical equation
Potassium	On gentle heating, it burns with lilac flame and give a white powder	Potassium + Oxygen → Potassium oxide $4K(s) + O_2(g) \rightarrow 2K_2O(s)$
Sodium	On gentle heating, it burns with golden yellow flame and give a white powder	Sodium + Oxygen → Sodium oxide $4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$
Calcium	On strong heating, it burns with brick-red flame and give a white powder	Calcium + Oxygen → Calcium oxide $2Ca(s) + O_2(g) \rightarrow 2CaO(s)$
Magnesium	On strong heating, it burns with bright white flame and give a white powder	Magnesium + Oxygen → Magnesium oxide $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
Aluminium	On strong heating, it burns to give out heat and a white powder	Aluminium + Oxygen → Aluminium oxide $4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$
Zinc	On strong heating, it burns to give out heat and a powder (Hot: Yellow; Cold: white)	Zinc + Oxygen → Zinc oxide $2Zn(s) + O_2(g) \rightarrow 2ZnO(s)$
Iron	On strong heating, it burns with yellow spark and give a black powder (Note: Iron(III) oxide is brown in color)	Iron + Oxygen → Iron(II, III) oxide $3Fe(s) + 2O_2(g) \rightarrow Fe_3O_4(s)$
Lead	On strong heating, it melts and gives a powder (Hot: Orange; Cold: yellow)	Lead + Oxygen → Lead(II) oxide $2Pb(s) + O_2(g) \rightarrow 2PbO(s)$
Copper	On very strong heating, it gives a black powder	Copper + Oxygen → Copper(II) oxide $2Cu(s) + O_2(g) \rightarrow 2CuO(s)$
Mercury	On very strong heating, it gives a red powder	Mercury + Oxygen → Mercury(II) oxide $2Hg(l) + O_2(g) \rightarrow 2HgO(s)$
Silver	No reaction with oxygen (Note: Silver oxide (Ag ₂ O) is black in color)	---
Platinum	No reaction with oxygen	---
Gold	No reaction with oxygen	---

- Potassium and sodium should be stored under **paraffin oil** (Shiny surface of reactive metals turns dull when exposed in air)
- Calcium should be stored in an **airtight container**

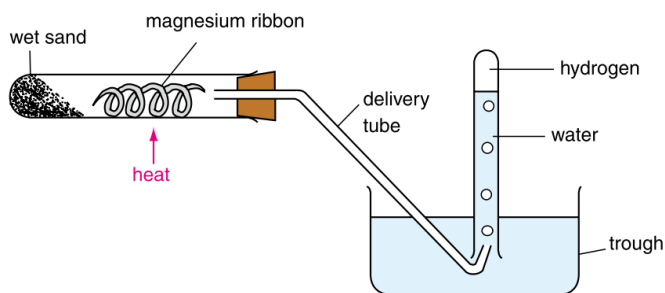
Intensive notes (Topic 3: Metals)

Reaction of metals with water / steam

Metal	Observation	Word equation / chemical equation
Potassium	Reacts with cold water : - Melts to form a silvery ball - Moves on water surface - Hissing sound is heard - Burns with lilac flame	Potassium + Water → Potassium hydroxide + Hydrogen $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$
Sodium	Reacts with cold water : - Melts to form a silvery ball - Moves on water surface - Hissing sound is heard - Burns with golden yellow flame	Sodium + Water → Sodium hydroxide + Hydrogen $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$
Calcium	Reacts with cold water : - Sinks to the bottom - Reacts readily to give colorless gas - Forms a milky suspension	Calcium + Water → Calcium hydroxide + Hydrogen $Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$
Magnesium	Reacts with hot water slowly . Reacts with steam vigorously : - Give an intense white light to form a white solid	Magnesium + <u>Water</u> → Magnesium <u>hydroxide</u> + Hydrogen $Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(aq) + H_2(g)$ Magnesium + <u>Steam</u> → Magnesium <u>oxide</u> + Hydrogen $Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$
Aluminium	Aluminium usually covered by a thin layer of aluminium oxide which prevents the reaction between aluminium and steam.	Aluminium + Steam → Aluminium oxide + Hydrogen $2Al(s) + 3H_2O(g) \rightarrow Al_2O_3(s) + 3H_2(g)$
Zinc	Reacts with steam to give a powder (Hot: Yellow; Cold: White)	Zinc + Steam → Zinc oxide + Hydrogen $Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_2(g)$
Iron	Reacts with steam to give a black powder	Iron + Steam → Iron(II, III) oxide + Hydrogen $3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$
Lead, Copper, Mercury, Silver, Platinum, Gold	No reaction with water or steam	---



Calcium + Water set-up



Magnesium + Water set-up

Intensive notes (Topic 3: Metals)

Reaction of metals with dilute sulphuric acid and dilute hydrochloric acid

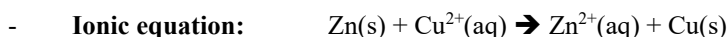
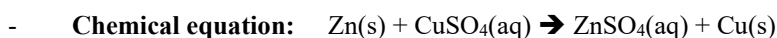
Metal	Observation	Word equation / chemical equation
Potassium Sodium	React explosively because flammable hydrogen gas is given off and the reaction produced a lot of heat	Not attempted as the reaction is explosive
Calcium	Reacts readily to give colorless gas *When the metal surface is fully covered by insoluble calcium sulphate, the reaction stops as the metal is no longer in contact with the acid.	Calcium + hydrochloric acid → Calcium chloride + hydrogen $\text{Ca(s)} + 2\text{HCl(aq)} \rightarrow \text{CaCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ Calcium + sulphuric acid → Calcium sulphate + hydrogen $\text{Ca(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{CaSO}_4\text{(s)} + \text{H}_2\text{(g)}$
Magnesium	Reacts readily to give colorless gas	Magnesium + hydrochloric acid → Magnesium chloride + hydrogen $\text{Mg(s)} + 2\text{HCl(aq)} \rightarrow \text{MgCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ Magnesium + sulphuric acid → Magnesium sulphate + hydrogen $\text{Mg(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{MgSO}_4\text{(aq)} + \text{H}_2\text{(g)}$
Aluminium	Reacts readily to give colorless gas	Aluminium + hydrochloric acid → Aluminium chloride + hydrogen $2\text{Al(s)} + 6\text{HCl(aq)} \rightarrow 2\text{AlCl}_3\text{(aq)} + 3\text{H}_2\text{(g)}$ Aluminium + sulphuric acid → Aluminium sulphate + hydrogen $2\text{Al(s)} + 3\text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{Al}_2\text{(SO}_4\text{)}_3\text{(aq)} + 3\text{H}_2\text{(g)}$
Zinc	Reacts readily to give colorless gas	Zinc + hydrochloric acid → Zinc chloride + hydrogen $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ Zinc + sulphuric acid → Zinc sulphate + hydrogen $\text{Zn(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{H}_2\text{(g)}$
Iron	Reacts readily to give colorless gas Solution turns from colorless to green	Iron + hydrochloric acid → Iron(II) chloride + hydrogen $\text{Fe(s)} + 2\text{HCl(aq)} \rightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ Iron + sulphuric acid → Iron(II) sulphate + hydrogen $\text{Fe(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{FeSO}_4\text{(aq)} + \text{H}_2\text{(g)}$
Lead	Reacts very slowly to give colorless gas * When the metal surface is fully covered by insoluble lead(II) sulphate / lead(II) chloride, the reaction stops as the metal is no longer in contact with the acid.	Lead + hydrochloric acid → Lead(II) chloride + Hydrogen $\text{Pb(s)} + 2\text{HCl(aq)} \rightarrow \text{PbCl}_2\text{(s)} + \text{H}_2\text{(g)}$ Lead + sulphuric acid → Lead(II) sulphate + Hydrogen $\text{Pb(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{PbSO}_4\text{(s)} + \text{H}_2\text{(g)}$
Copper, Mercury, Silver, Platinum, Gold	No reaction with both acids	---

Displacement reaction

Displacement reaction is a reaction in which one element displaces another element from its compound.

Example 1: Zinc + Copper(II) sulphate

- Zinc is more reactive than copper. Zinc displaces copper metal from the copper(II) sulphate solution.

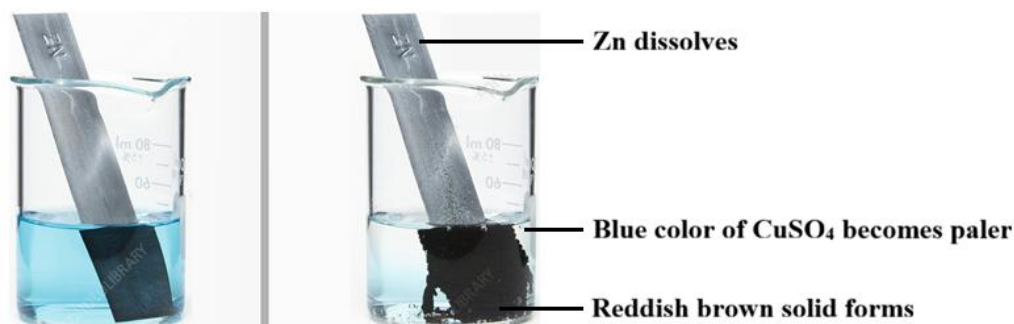


Observation 1: Zinc dissolves (Zinc metal loses electrons to form Zn^{2+} ions which dissolves in water)

Observation 2: Color intensity of blue $\text{CuSO}_4\text{(aq)}$ decreases (Cu^{2+} ions gains electrons to form copper metal)

Observation 3: Reddish brown solid forms (Cu^{2+} ions gains electrons to form copper metal)

Observation 4: Heat evolves (Displacement is an exothermic reaction)

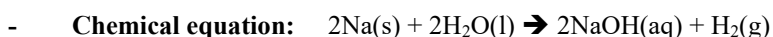


Example 2: Silver + Zinc sulphate

- Silver is less reactive than zinc. There is no reaction between silver and zinc sulphate solution

Example 3: Sodium + Copper(II) sulphate

- For metals that can react with water (e.g. sodium, potassium, calcium), they will **react with water directly** instead of the metal ions dissolved to give metal hydroxide and hydrogen gas



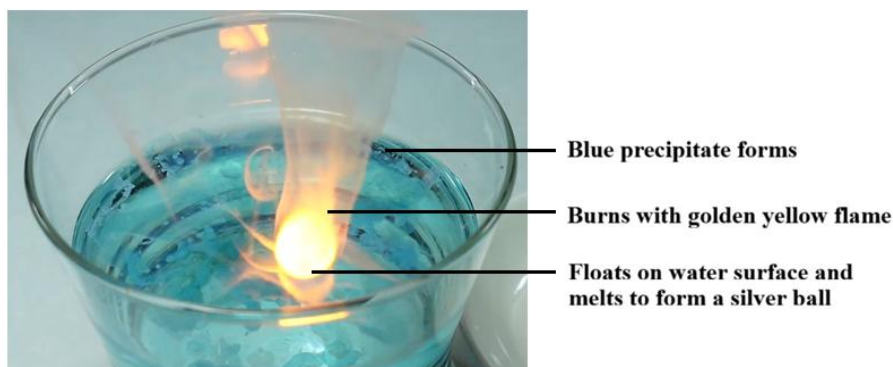
Observation 1*: Sodium burns with golden yellow flame

Observation 2*: Sodium floats on water surface and melts to form a silvery ball

Observation 3*: Hissing sound is heard

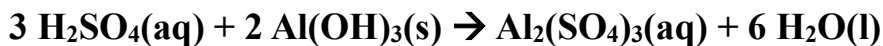
Observation 4: Blue precipitate forms ($\text{Cu}^{2+}\text{(aq)}$ reacts with $\text{OH}^-\text{(aq)}$ to form $\text{Cu(OH)}_2\text{(s)}$ which is insoluble in water)

*Same observations for the reaction between sodium and water

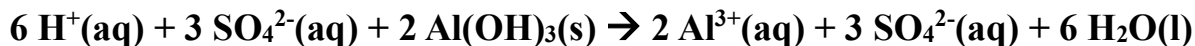


Steps for writing an ionic equation

1. Write down the FULL chemical equation



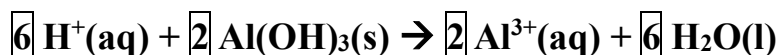
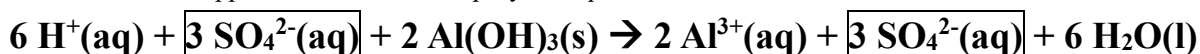
2. Split the compounds that are soluble in water (aqueous state) into individual ions



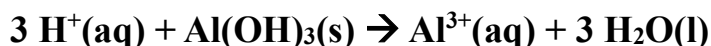
Note: Pay attention to the coefficient of H^+ . Do not write 3H_2 . It is an acid not hydrogen gas.

$\text{Al}(\text{OH})_3$ is insoluble in water. Do NOT split it into Al^{3+} and 3OH^-

3. Cancel out the ions that appear in both sides. Simplify the equation if needed.

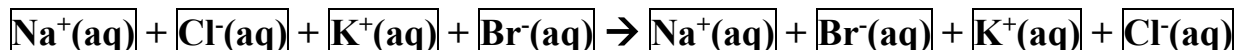


Ionic equation:



Note: Since SO_4^{2-} does not participate in the reaction, we call them 'spectator ion'.

4. If ALL ions are cancelled out, there is NO REACTION between 2 compounds



States of common chemicals

Gas (g)	Liquid (l)
Noble gases, F_2 , Cl_2 , O_2 , N_2 , H_2 , CO_2 , CO , NO_2 , NO , SO_3 , SO_2 , CH_4 , NH_3 , H_2S , PH_3	Br_2 , Hg , H_2O , CH_3OH
Solid (s)	Aqueous solution (aq)
All Insoluble salts, All metals (except Hg) C , P_4 , S_8 , SiO_2 , I_2 , B , Si	Substances that are soluble in water (soluble salts, acids, alkalis) $\text{Cl}_2(\text{g})$, $\text{Br}_2(\text{l})$, and $\text{I}_2(\text{s})$ can exist as $\text{Cl}_2(\text{aq})$, $\text{Br}_2(\text{aq})$ and $\text{I}_2(\text{aq})$

Intensive notes (Topic 3: Metals)

Number of particle calculation

$$\text{Mole} = \frac{\text{Mass}}{\text{Molar mass}}$$

$$(\text{Mole}) \times (\text{Avogadro's constant}) = \text{No. of formula unit}$$

- Calculate the number of oxygen atoms in 60.1 g of quartz. (Relative atomic masses: O = 16.0, Si = 28.1)
 Answer: Mole of $\text{SiO}_2 = (\text{mass}) / (\text{molar mass}) = 60.1 / (28.1 + 16 \times 2) = 1$
 No. of formula unit (SiO_2) = (mole) $\times$ (Avogadro's constant) = $1 \times 6.02 \times 10^{23} = 6.02 \times 10^{23}$
 Since 1 SiO_2 contains **2 oxygen atoms**, number of oxygen atoms = $6.02 \times 10^{23} \times 2 = 1.204 \times 10^{24}$
- Calculate the number of atoms in 14.0 g of nitrogen gas. (Relative atomic mass of N = 14.0)
 Answer: Mole of nitrogen (N_2) = mass / molar mass = $14 / 14 \times 2 = 0.5$
 No. of formula unit (N_2) = (mole) $\times$ (Avogadro's constant) = $0.5 \times (6.02 \times 10^{23}) = 3.01 \times 10^{23}$
 Since 1 N_2 molecule contains **2 N atoms**, number of atoms = $3.01 \times 10^{23} \times 2 = 6.02 \times 10^{23}$
- Calculate the number of ions in 1 mol of iron(III) sulphate
 Answer: No. of formula unit ($\text{Fe}_2(\text{SO}_4)_3$) = (mole) $\times$ (Avogadro's constant) = $1 \times (6.02 \times 10^{23}) = 6.02 \times 10^{23}$
 Since 1 $\text{Fe}_2(\text{SO}_4)_3$ contains **2 Fe^{3+} ions** and **3 SO_4^{2-} ions**, number of ions = $6.02 \times 10^{23} \times (2+3) = 3.01 \times 10^{24}$ ions
- Calculate the mass of 534202 oxygen molecules. (Relative atomic mass of O = 16.0)
 Answer: No. of moles of oxygen molecules (O_2) = (no. of formula unit) / (Avogadro's constant)
 $= 534202 / 6.02 \times 10^{23} = 8.87 \times 10^{-19}$
 Mass of 534202 oxygen molecules = (mole) $\times$ (molar mass) = $8.87 \times 10^{-19} \times (16.0 \times 2) = 2.84 \times 10^{-17}$ g.

Percentage by mass calculation

- Calculate the mass of each element in 2.0 g $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. (Relative atomic masses: H = 1.0, O = 16.0, S = 32.1, Cu = 63.5)
 Formula mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O} = 63.5 + 32.1 + (4)(16) + (5)(1 \times 2 + 16) = 249.6$

	Cu	S	O	H
% by mass	$63.5/249.6 = 25.4\%$	$32.1/249.6 = 12.9\%$	$(16)(4+5)/249.6 = 57.7\%$	$(1)(5 \times 2)/249.6 = 4.0\%$
Mass of element	$2.0 \times 25.4\% = 0.51$ g	$2.0 \times 12.9\% = 0.26$ g	$2.0 \times 57.7\% = 1.15$ g	$2.0 \times 4.0\% = 0.08$ g

- Hydrated salt $\text{X} \cdot n\text{H}_2\text{O}$ contains 51.6 % of water by mass. Given that the molar mass of X is 120.3 g, what is n?
 (Relative atomic masses : H = 1.0, O = 16.0)
 Answer (mass of water) / (mass of hydrated salt) = $[(18)(n)] / [120.3 + (18)(n)] = 51.6\%$
 $n = 7$ (n must be an integer)
- When 3.80 g of the hydrated magnesium sulphate crystals ($\text{MgSO}_4 \cdot x\text{H}_2\text{O}$) are heated, 2.00 g of anhydrous magnesium sulphate are produced. What is the value of x? (Formula masses: $\text{MgSO}_4 = 120.4$, $\text{H}_2\text{O} = 18.0$)
 Answer (mass of anhydrous salt) / (mass of hydrated salt) = $(2.00) / (3.80) = (120.4) / (120.4 + 18x)$
 $x = 6$ (x must be an integer)

Intensive notes (Topic 3: Metals)

Deduce the empirical formula and molecular formula of a compound

1. An oxide of lead contains 90.6% of lead by mass. Calculate the empirical formula of the oxide.

	Pb (R.A.M. = 207.2)	O (R.A.M. = 16.0)
Mass ratio (in 100 g)	90.6	100 – 90.6 = 9.4
Mole ratio	90.6 / 207.2 = 0.4373	9.4 / 16.0 = 0.5875
Simplest ratio*	0.4373/0.4373 = 1	0.5875/0.4373 = 1.34

*Since 1 : 1.34 = (1x3) : (1.34x3) = 3 : 4, the empirical formula of the oxide is Pb₃O₄

2. 23.90 g of iron is allowed to react with excess oxygen until it is completely oxidized. The mass of metal oxide obtained is 34.18 g. What is the empirical formula of the oxide?

	Fe (R.A.M. = 55.8)	O (R.A.M. = 16.0)
Mass ratio	23.9	34.18 – 23.90 = 10.28
Mole ratio	23.9 / 55.8 = 0.4283	10.28 / 16.0 = 0.6425
Simplest ratio*	0.4283/0.4283 = 1	0.6425/0.4283 = 1.5

*Since 1 : 1.5 = 2 : 3, the formula of the oxide is Fe₂O₃

3. Compound X is an oxide of nitrogen. X contains 30.43% of nitrogen by mass. If the molecular mass of X is 92, deduce the molecular formula of X.

	N (R.A.M. = 14.0)	O (R.A.M. = 16.0)
Mass ratio (in 100 g)	30.43	69.57
Mole ratio	30.43 / 14.0 = 2.174	69.57 / 16.0 = 4.348
Simplest ratio	2.174 / 2.174 = 1	4.348 / 2.174 = 2

The empirical formula of X is NO₂. Let (NO₂)_n be the molecular formula of X

$$n(14 + 16 \times 2) = 92$$

n = 2 (Molecular formula of X is N₂O₄)

4. Compound W contains carbon, hydrogen and oxygen only. The relative molecular mass of W is 88. Complete combustion of 1.32 g of W gives 2.64 g of carbon dioxide and 1.08 g of water. Deduce the molecular formula of W.

Note: All the C in CO₂ and H in H₂O come from W

	C (R.A.M. = 12.0)	H (R.A.M. = 1.0)	O (R.A.M. = 16.0)
Mass	(Mass of CO ₂)(% by mass of C) = 2.64 x [12 / (12 + 16x2)] = 0.72	(Mass of H ₂ O)(% by mass of H) = 1.08 x [1x2 / (1x2 + 16)] = 0.12	(Mass of W) – (C) – (H) = 1.32 – 0.72 – 0.12 = 0.48
Mole	0.72 / 12.0 = 0.06	0.12 / 1.0 = 0.12	0.48 / 16.0 = 0.03
Simplest ratio	0.06 / 0.03 = 2	0.12 / 0.03 = 4	0.03 / 0.03 = 1

The empirical formula is C₂H₄O. Let the molecular formula be (C₂H₄O)_n

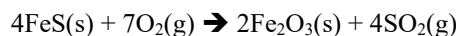
$$n(12 \times 2 + 1 \times 4 + 16) = 88$$

n = 2 (The molecular formula is C₄H₈O₂)

Reacting masses and percentage yield

$$\text{Percentage yield} = \frac{\text{Actual yield}}{\text{Theoretical yield}}$$

The equation for the reaction between FeS and O₂ is shown below: (Relative atomic masses: O = 16.0, S = 32.1, Fe = 55.8)



1. In an experiment, 10.0 g of FeS is allowed to react with excess O₂.

(a) Calculate the theoretical yield of Fe₂O₃.

Answer: $n_{\text{FeS used}} = (\text{mass}) / (\text{molar mass}) = 10.0 / (55.8 + 32.1) = 0.1138$

Theoretical $n_{\text{Fe}_2\text{O}_3 \text{ formed}} = (n_{\text{FeS}}) \times 2/4 = 0.1138 \times 2/4 = 0.05688$

Theoretical mass of Fe₂O₃ formed = $(n_{\text{Fe}_2\text{O}_3})(\text{molar mass}) = 0.05688 \times (55.8 \times 2 + 16 \times 3) = 9.08 \text{ g}$

(b) If the actual mass of Fe₂O₃ obtained is 6.3 g, calculate the percentage yield of Fe₂O₃.

Answer: $\text{Percentage yield} = (\text{Actual yield}) / (\text{Theoretical yield}) \times 100\% = 6.3/9.08 \times 100\% = 69.4\%$

(c) Suggest a reason why the percentage yield cannot reach 100%.

Answer: Incomplete reaction / there are side reactions / some product is lost in some of the experimental processes, such as purification / impurities are present in the reactants

2. In another experiment, 15.0 g of FeS is allowed to react with 10.0 g of O₂.

(a) Calculate the theoretical yield of Fe₂O₃.

Answer: (Since the amounts of both reactants are specified, we need to find out the limiting reactant first)

$n_{\text{FeS used}} = (\text{mass}) / (\text{molar mass}) = 15.0 / (55.8 + 32.1) = 0.1706$

$n_{\text{O}_2 \text{ needed to completely reaction with } 0.1706 \text{ mol of FeS}} = (n_{\text{FeS}}) \times 7/4 = 0.1706 \times 7/4 = \mathbf{0.2986 \text{ mol}}$

$n_{\text{O}_2 \text{ available}} = (\text{mass}) / (\text{molar mass}) = 10.0 / 32 = \mathbf{0.3125 \text{ mol} > 0.2986 \text{ mol}}$

Therefore O₂ is in excess while FeS is the **limiting reactant**

(To find out the theoretical $n_{\text{Fe}_2\text{O}_3}$ formed, we should calculate from the limiting reactant, not the excess)

Theoretical $n_{\text{Fe}_2\text{O}_3} = (n_{\text{FeS}}) \times 2/4 = 0.1706 \times 2/4 = 0.08532$

Theoretical mass of Fe₂O₃ = $(n_{\text{Fe}_2\text{O}_3})(\text{molar mass}) = 0.08532 \times (55.8 \times 2 + 16 \times 3) = 13.6 \text{ g}$

(b) If the percentage yield of Fe₂O₃ is 52.0%, calculate the actual yield of Fe₂O₃ obtained.

Answer: $\text{Percentage yield} = (\text{Actual yield}) / (\text{Theoretical yield}) \times 100\%$

$52.0\% = (\text{Actual yield}) / (13.6)$

Actual yield = 7.08 g

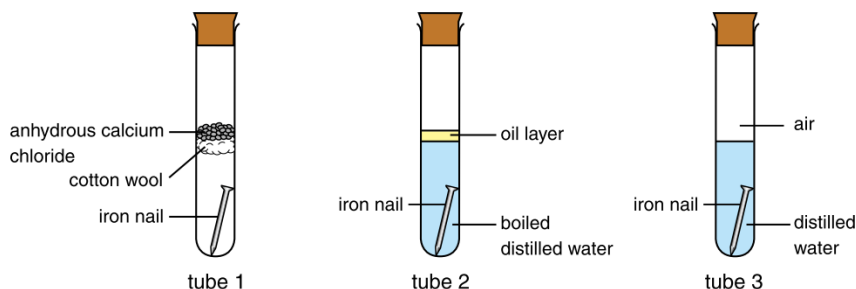
Corrosion

- Corrosion of metal refers to the gradual deterioration of a metal, resulting from the reaction of the metal with air, water or other substances in the environment.
- In general, the more reactive a metal, the more rapidly it corrodes.

Note: Rusting refers to the corrosion of **iron** only, but not other metals

Essential conditions for rusting

Essential conditions of rusting: Water and oxygen



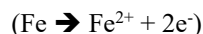
Tube 1: No rusting will occur. Anhydrous calcium chloride is a drying agent. It removes water from the air. Therefore the iron nail in tube 1 has no contact with water.

Tube 2: No rusting will occur. Boiled distilled water removes dissolved oxygen in water. The oil layer prevents oxygen from dissolving in water again. Therefore the iron nail in tube 2 has no contact with oxygen.

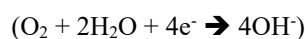
Tube 3: Rusting will occur. It is because iron nail in tube 3 has contact with water and oxygen. Note that normal distilled water still contains dissolved oxygen.

Chemistry of rusting

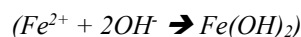
Step 1: Iron $\rightarrow$ Iron(II) ion



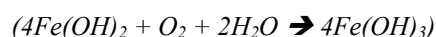
Water + Oxygen $\rightarrow$ Hydroxide ion



Step 2: Iron(II) ion + Hydroxide ion $\rightarrow$ Iron(II) hydroxide



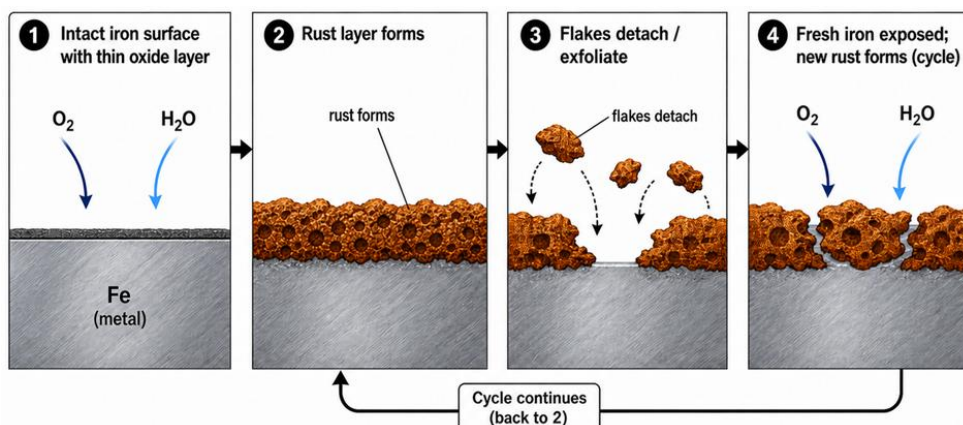
Step 3: Iron(II) hydroxide + Water + Oxygen $\rightarrow$ Iron(III) hydroxide



Step 4: Iron(III) hydroxide $\rightarrow$ Hydrated iron(III) oxide (Rust, reddish brown solid)

Overall equation: $4\text{Fe}(\text{s}) + 3\text{O}_2(\text{g}) + 2\text{nH}_2\text{O}(\text{l}) \rightarrow 2\text{Fe}_2\text{O}_3 \cdot \text{nH}_2\text{O}$

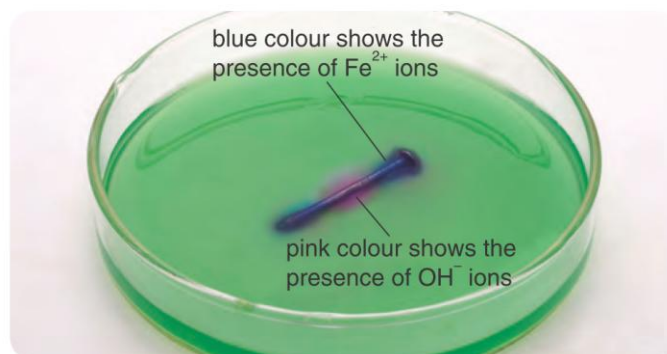
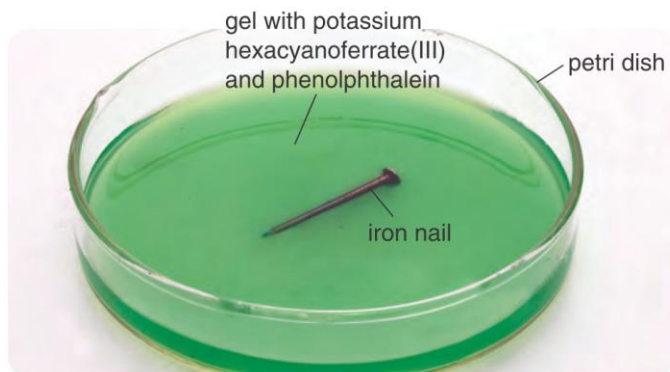
The rust formed is **loosely attached** to the iron surface. It **flakes off** from the iron surface easily. When the inner fresh iron surface is exposed to air, it can further react with oxygen and water. Rusting continues until the iron piece corrodes completely.



Rust indicator

Components in rust indicator:

1. Potassium hexacyanoferrate(III) ($\text{K}_3\text{Fe}(\text{CN})_6$), turns **blue** in the presence of Fe^{2+} ions.
2. Phenolphthalein, turns **pink** in the presence of OH^- ions.
3. Sodium chloride, added to speed up rusting



Blue area: When iron rusts, iron(II) ion (Fe^{2+}) will be formed. It turns **potassium hexacyanoferrate(III) ($\text{K}_3\text{Fe}(\text{CN})_6$) blue.**

Pink area: Hydroxide ion (OH^-) forms during rusting. It turns **phenolphthalein pink.**

Note 1: Uneven, scratched, bent or sharp area of an iron object will rust more readily.

Note 2: Gel can provide a medium for the movement of ions but to prevent the spread of the colour

Factors that speed up rusting

1. **Presence of soluble ionic compounds**
 - Salt contains **mobile ions** that can increase the electrical conductivity and hence speed up the formation of ions.
2. **Presence of acid**
 - Acid contains **mobile ions** that can increase the electrical conductivity and hence speed up the formation of ions.
 - Acid promotes iron(II) ion formation ($\text{Fe} + 2\text{H}^+ \rightarrow \text{Fe}^{2+} + \text{H}_2$)
3. **Presence of metal ions that can undergo displacement reaction with iron**
 - These metal ions contain **mobile ions** that can increase the electrical conductivity and hence speed up the formation of ions.
 - These metal ions also promote iron(II) ion formation (e.g. $\text{Fe} + 2\text{Pb}^{2+} \rightarrow \text{Fe}^{2+} + \text{Pb}$)
4. **High temperature**
 - Higher temperature, higher rate of chemical reactions
5. **Attachment of less reactive metal (e.g. tin, lead, copper, silver) to iron**
 - Less reactive metal, when connected to iron, causes iron to lose electrons more readily.
 - On the contrary, when iron is attached to a more reactive metal, rusting would not occur. This is because iron is protected by sacrificial protection

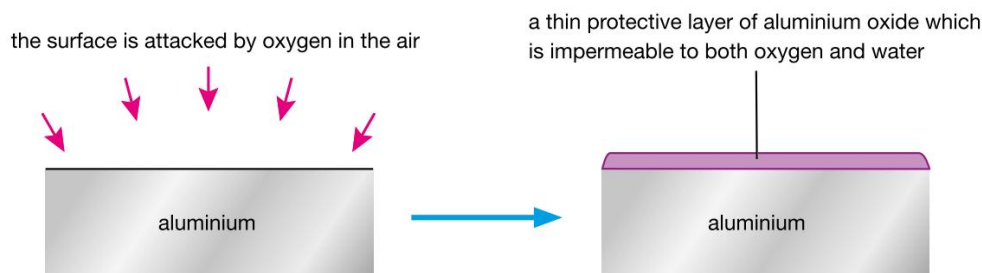
Intensive notes (Topic 3: Metals)

Prevention of rusting

Method of rust prevention	Chemistry	Advantage	Disadvantage	Example
Painting	Added layer prevents the iron object from contact with air and water	- Cheap	- Scratched off	1. Bridges 2. Ships 3. Fences 4. Car bodies
Coating with plastic		- Lasts long - Looks good	- More expensive than painting	1. Coat hangers 2. Paper clips
Greasing		- Does not fall off - Lubricating effect	- Not once and for all - Dirt would stick to grease	1. Moving parts of machine 2. Woodworking tools
Galvanizing (zinc-plating)		- In case the zinc is scratched, the iron is still protected	- Zinc ions are toxic	1. Construction 2. Buckets
Tin-plating		- Tin is corrosion resistant - Tin is not toxic	- When tin is scratched, iron rusts more quickly as tin is less reactive than iron	1. Tin food cans
Electroplating		- Beautiful appearance	- Expensive	1. Water taps 2. Car bumpers
Cathodic protection	Iron object is connected to the negative terminal of an electric source. It supplies electrons to prevent iron from losing electrons	- Convenient	- Not applicable to many objects	1. Car bodies 2. Underground water pipes 3. Storage tanks 4. Steel pier legs
Sacrificial protection	Iron object is connected to a more reactive metal. It loses electrons more readily than iron.	- Effective	- Sacrificed metal needs replacement	1. Galvanized iron 2. Pipelines connected to Mg
Alloying	Iron object is mixed uniformly with other elements such as C, Cr, Mn. (Cr_2O_3 layer which is impermeable to O_2 and H_2O)	- Beautiful appearance - Effective	- Expensive	1. Cookware 2. Cutlery

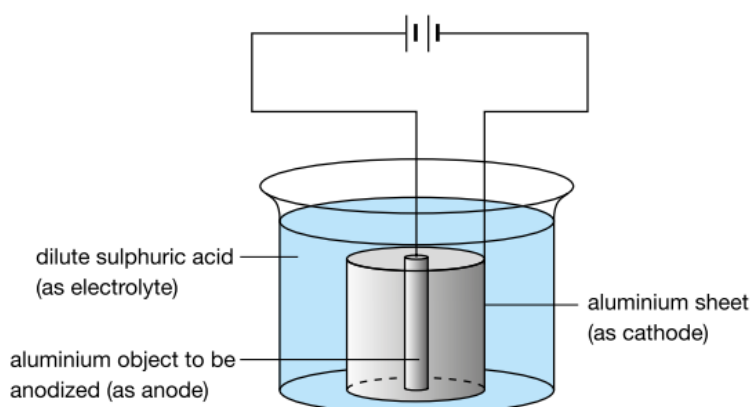
Corrosion resistance of aluminium

When aluminium is exposed in air, it reacts with oxygen in air immediately to form a thin layer of aluminium oxide (which adheres firmly on the surface of aluminium metal) which is impermeable to water and oxygen.



Anodization: A process to thicken the aluminium oxide layer on the surface of aluminium.

In anodization, the aluminium object is connected to a **positive terminal** of the battery



Half equation for the change at anode (+ve electrode)	Half equation for the change at cathode (-ve electrode)
$2\text{Al(s)} + 3\text{H}_2\text{O(l)} \rightarrow \text{Al}_2\text{O}_3\text{(s)} + 6\text{H}^+\text{(aq)} + 6\text{e}^-$	$2\text{H}^+\text{(aq)} + 2\text{e}^- \rightarrow \text{H}_2\text{(g)}$

Overall reaction of anodization: $2\text{Al(s)} + 3\text{H}_2\text{O(l)} \rightarrow \text{Al}_2\text{O}_3\text{(s)} + 3\text{H}_2\text{(g)}$

Anodized aluminium object has a **higher hardness**, **higher corrosion resistance** and can be **dyed more easily**.